phylogenetic examples

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2022-10-12

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This document contains the code to replicate the two analyses of macaque hierarchy steepness (one analysis in the main text, one in the appendix) from *Extending Bayesian Elo-rating to quantify the steepness of dominance hierarchies* (https://doi.org/10.1101/2022.01.28.478016). It was built using `EloSteepness` (version 0.4.6) and `EloSteepness.data` (version 0.9.6). Models in this document were fitted using `cmdstanr` (version 0.5.2) with Stan (v. 2.30.1).

Please note that in earlier iterations of this document and the accompanying preprint, we modeled steepness as beta distributed, as opposed to normally distributed. For completeness sake, the `Stan` code for those earlier models is still included in the package and we point out the necessary modifications to the `R` code below to fit them.

A second thing to note is that we provide two versions of Stan files. These differ by whether they use the array syntax for declaring data structures (see here for details). The code in the Stan files as referenced and used in this document relies on the new syntax and should work fine with recent versions of Stan used through `cmdstanr` (Stan v. > 2.26), but might fail if an outdated version of Stan is used. The alternative versions are installed in the same folder as the used ones, with `_oldsyntax` appended to the file name.

```
library(EloSteepness)
library(EloSteepness.data)
library(rstan)
library(cmdstanr)
library(ape)
library(xtable) # required only for table generation in this document
library(brms) # required only for one comparison model
```

1 Comparative example from the main text

First provide the path the to the Stan model code[1].

```
# path to raw model code
mod1 <- system.file("extdata/phylogenetic_examples/comparative_model_1_gaussian.stan",
                    package = "EloSteepness.data")
file.exists(mod1)
#> [1] TRUE
```

Next, compile the model.

```
# compile stan model
xmod1 <- cmdstan_model(stan_file = mod1, compile = TRUE)
```

The raw Stan code can be inspected like this:

```
xmod1$print()
```

The next chunks of code deal with data processing and preparation.

```
# load tree, matrices and key table
data("Primates301_nex")
data("phylo_matrices")
data("phylo_key")

# prune tree to available species
species <- unique(phylo_key$species_fixed)
tree <- keep.tip(Primates301_nex, species)
# plot(tree)
length(tree$tip.label) == length(species) # sanity check
#> [1] TRUE

# sort matrices and key so they have the same order
interaction_matrices <- phylo_matrices[sort(names(phylo_matrices))]
key <- phylo_key[order(phylo_key$datasetname), ]
all(names(interaction_matrices) == key$datasetname) # sanity check
#> [1] TRUE

# summary data for each group
specdata <- data.frame(datasetname = key$datasetname,
                       spec = gsub(" ", "_", key$species_fixed),
                       groupsize = NA,
                       n_interactions = NA,
                       dyads = NA,
                       prunks = NA)

for (m in seq_len(nrow(specdata))) {
  mat <- interaction_matrices[[specdata$datasetname[m]]]
  specdata$groupsize[m] <- ncol(mat)
  specdata$n_interactions[m] <- sum(mat)
  specdata$dyads[m] <- ncol(mat) * (ncol(mat) - 1) / 2
  specdata$prunks[m] <- EloRating::prunk(mat)[1]
}
```

For the sake of this vignette we reduce the number of species and also the number of interactions in the

[1] Replace "comparative_model_1_gaussian.stan" with "comparative_model_1.stan" to load and fit the original beta model.

networks. We do this just because it speeds up the sampling from the model. With the full data set, this would take up to a few hours (about two on my fairly old laptop[2]). With the reduced data set this takes less time to run (but still might take a few minutes).

```
# ignore these lines if you want to run the full data set
specdata <- specdata[specdata$groupsize < 7, ]
interaction_matrices <- interaction_matrices[specdata$datasetname]
tree <- keep.tip(tree, specdata$spec)
interaction_matrices <- lapply(interaction_matrices, function(x) ceiling(x / 5))
```

Now prepare the data to pass it to `Stan`.

```
# prepare the data set as list for sending it to Stan
xdata <- list()
xdata$n_datasets <- length(interaction_matrices)
xdata$n_ids_per_dataset <- as.integer(unlist(lapply(interaction_matrices, ncol)))
xdata$n_total_ids <- sum(xdata$n_ids_per_dataset)
xdata$n_interactions_per_dataset <- as.integer(unlist(lapply(interaction_matrices, sum)))
xdata$n_total_interactions <- sum(xdata$n_interactions_per_dataset)
xdata$interaction_index <- as.integer(seq_len(xdata$n_total_interactions))
xdata$individual_index <- as.integer(seq_len(xdata$n_total_ids))

# generate sequences
set.seed(1)
s <- lapply(interaction_matrices, randomized_sequence_from_matrix)
# global winner and loser indices
xdata$winner <- do.call("rbind", lapply(s, function(x)x$winnermat))
xdata$loser <- do.call("rbind", lapply(s, function(x)x$losermat))

xdata$index_dataset_interactions_start <- cumsum(xdata$n_interactions_per_dataset) -
                                          xdata$n_interactions_per_dataset + 1
xdata$index_individuals_start <- cumsum(xdata$n_ids_per_dataset) -
                                 xdata$n_ids_per_dataset + 1

xdata$y <- rep(1, xdata$n_total_interactions)

# phylogeny-related
n_spec <- length(tree$tip.label)
tree_mat <- vcv.phylo(tree, corr = TRUE)
species_index <- integer(nrow(specdata))
for (i in seq_len(nrow(specdata))) {
  species_index[i] <- which(colnames(tree_mat) == specdata$spec[i])
}

# for phylogeny part of the model
xdata$N_phyl <- n_spec # N_1
xdata$N_coeff_phyl <- 1 # number of parameters: 1 = SD for phyl 'intercepts'
xdata$spec_index <- species_index
xdata$chol_mat <- t(chol(tree_mat)) # Cholesky factor of phylogenetic correlation matrix
xdata$phyl_predictor <- rep(1, nrow(specdata)) # group-level predictor values

# for repeated measures part of the model
xdata$N_repeated <- n_spec # the same as for the phylogeny part
# number of parameters: 1 = SD for repeated measurements (species intercepts)
```

[2] and about 4 hours with the original beta model...

```
xdata$N_coeff_repeated <- 1
xdata$index_rep_measures <- species_index # the same as for the phyl part
# group-level predictor values, the same as for the phyl part
xdata$rep_measures_predictor <- rep(1, nrow(specdata))
```

Now we can sample from the actual model.

```
# sampling
res1 <- xmod1$sample(data = xdata,
                     refresh = 100,
                     parallel_chains = 2,
                     iter_warmup = 1500,
                     iter_sampling = 2500,
                     seed = 1,
                     adapt_delta = 0.9)
```

And then inspect the results[3].

```
stanfit <- rstan::read_stan_csv(res1$output_files())
xsummary <- summary(stanfit)$summary
round(summary(stanfit, pars = c("sd_repeated_measures", "sd_phyl", "sd_residual"))$summary, 2)
#>                          mean se_mean   sd 2.5%  25%  50%  75% 97.5%   n_eff
#> sd_repeated_measures[1] 0.19    0.00 0.16 0.01 0.07 0.15 0.27  0.57 1288.70
#> sd_phyl[1]              0.23    0.01 0.20 0.01 0.08 0.17 0.31  0.75  526.84
#> sd_residual             0.51    0.00 0.13 0.30 0.42 0.50 0.59  0.80 1053.24
#>                          Rhat
#> sd_repeated_measures[1]    1
#> sd_phyl[1]                 1
#> sd_residual                1

# optional saving of results
# save(xmod1, xsummary, res1, stanfit, xdata, specdata, tree, file = "comp_analysis_1.RData")
```

```
par(mfrow = c(1, 2), family = "serif")
par(family = "serif", mgp = c(1.8, 0.7, 0), las = 1, mar = c(2, 1, 1, 1))

plot.phylo(tree, cex = 0.8, show.tip.label = FALSE, x.lim = c(0, 19))
tips <- tree$tip.label
for (i in seq_along(tips)) {
  x <- sum(specdata$spec == tips[i])
  y <- gsub("_", " ", tips[i])
  tiplabels(text = bquote(italic(.(y))~.(paste0(" (", x, ")"))),
            tip = i, bg = NULL, adj = 0, frame = "none", offset = 0.3, cex = 0.8)
}

par(mar = c(3, 2.5, 1.5, 1))

# posteriors
x <- as.numeric(extract(stanfit, pars = c("sd_phyl"))$sd_phyl)
y <- as.numeric(extract(stanfit, pars = c("sd_repeated_measures"))$sd_repeated_measures)
z <- as.numeric(extract(stanfit, pars = c("sd_residual"))$sd_residual) # ignore for beta model

# and their densities
p1 <- density(x, adjust = 1)
```

[3]Leave out (remove) `"sd_residual"` when fitting the original beta model.

```r
p2 <- density(y, adjust = 1)
p3 <- density(z, adjust = 1) # ignore for beta model


# priors
pr_phyl <- as.numeric(extract(stanfit, pars = c("prior_sd_phyl"))$prior_sd_phyl)
pr_rep <- as.numeric(extract(stanfit, pars = c("prior_sd_repeated_measures"))$prior_sd_repeated_measures
pr_resid <- as.numeric(extract(stanfit, pars = c("prior_sd_residual"))$prior_sd_residual) # ignore for

# and their densities
pr_phyl <- density(pr_phyl[pr_phyl < 10], adjust = 1) # truncated for smoother display
pr_rep <- density(pr_rep[pr_rep < 10], adjust = 1)
pr_resid <- density(pr_resid[pr_resid < 10], adjust = 1) # ignore for beta model

# plotting dimensions
yr <- c(0, max(p1$y, p2$y, p3$y) * 1.05) # ignore for beta model
# yr <- c(0, max(p1$y, p2$y) * 1.05) # use for beta model
xr <- c(0, max(p1$x, p2$x, p3$x) * 1.05) # ignore for beta model
# xr <- c(0, max(p1$x, p2$x) * 1.05) # use for beta model
xr[2] <- ceiling(xr[2])

# set up empty plot
plot(0, 0, type = "n", xlim = xr, ylim = yr, xaxs = "i", yaxs = "i", las = 1,
     axes = FALSE, xlab = "SD", ylab = "density")
axis(1)

# fill it
polygon(c(p1$x, rev(p1$x)), c(rep(0, length(p1$y)), rev(p1$y)),
        border = NA, col = adjustcolor("#3B99B1", 0.5))
polygon(c(p2$x, rev(p2$x)), c(rep(0, length(p2$y)), rev(p2$y)),
        border = NA, col = adjustcolor("#EACB2B", 0.5))
polygon(c(p3$x, rev(p3$x)), c(rep(0, length(p3$y)), rev(p3$y)),
        border = NA, col = adjustcolor("#F5191C", 0.5)) # ignore for beta model

polygon(c(pr_phyl$x, rev(pr_phyl$x)), c(rep(0, length(pr_phyl$y)), rev(pr_phyl$y)),
        border = NA, col = adjustcolor("grey", 0.4))
polygon(c(pr_rep$x, rev(pr_rep$x)), c(rep(0, length(pr_rep$y)), rev(pr_rep$y)),
        border = NA, col = adjustcolor("grey", 0.4))
polygon(c(pr_resid$x, rev(pr_resid$x)), c(rep(0, length(pr_resid$y)), rev(pr_resid$y)),
        border = NA, col = adjustcolor("grey", 0.4)) # ignore for beta model

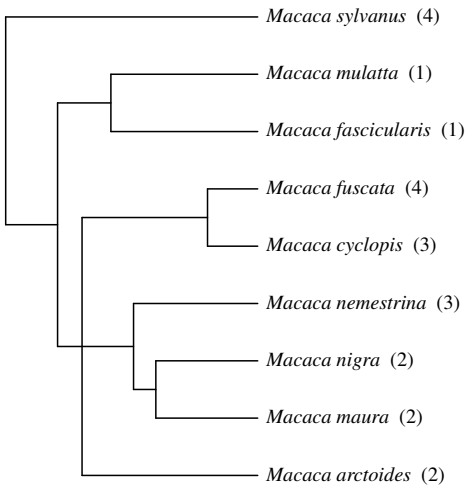

points(p1$x, p1$y, type = "l", lwd = 0.5)
points(p2$x, p2$y, type = "l", lwd = 0.5)
points(p3$x, p3$y, type = "l", lwd = 0.5) # ignore for beta model

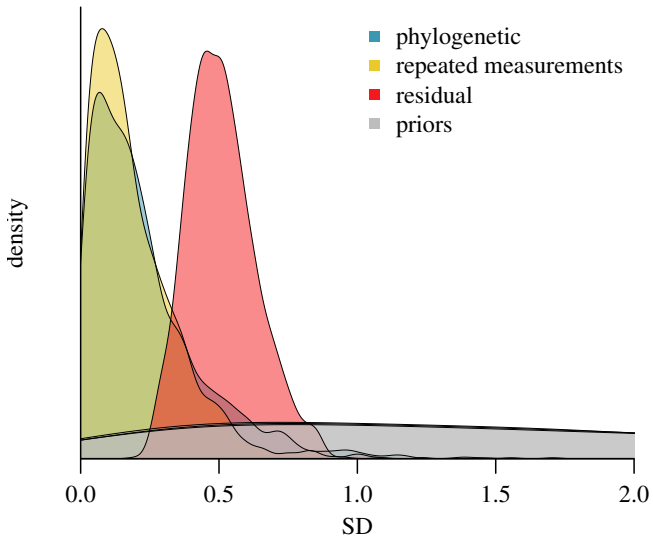
points(pr_phyl$x, pr_phyl$y, type = "l", lwd = 0.5)
points(pr_rep$x, pr_rep$y, type = "l", lwd = 0.5)
points(pr_resid$x, pr_resid$y, type = "l", lwd = 0.5) # ignore for beta model
box(bty = "l")


ldat <- matrix(c(
```

```
  "#3B99B1", "phylogenetic",
  "#EACB2B", "repeated measurements",
  "#F5191C", "residual", # comment/ignore this line for beta model
  "grey", "priors"
), ncol = 2, byrow = TRUE)

legend("topright", ncol = 1,
       col = ldat[, 1],
       legend = ldat[, 2],
       pch = 15, bty = "n")
```





2 Reanalysis of Balasubramaniam et al data set

The code to replicate this analysis requires a bit less data pre-processing than the model above. Most crucially, we don't start from the original matrices since we don't have access to all the raw data. Rather, we use the steepness values as they are reported in the original publication and fit a simpler model[4], i.e., without the actually applying the STEER algorithm.

```
# path to raw model code
mod2 <- system.file("extdata/phylogenetic_examples/comparative_model_2_gaussian.stan",
                    package = "EloSteepness.data")
file.exists(mod2)
#> [1] TRUE
```

```
# compile stan model
xmod2 <- cmdstan_model(stan_file = mod2, compile = TRUE)
```

```
# tree data
data("Primates301_nex")

# data from Balasubramaniam et al 2012 (table 2)
sp <- c("Macaca_assamensis", "Macaca_fascicularis", "Macaca_fascicularis",
        "Macaca_fuscata", "Macaca_fuscata", "Macaca_mulatta", "Macaca_mulatta",
        "Macaca_nigra", "Macaca_radiata", "Macaca_sylvanus", "Macaca_thibetana",
        "Macaca_thibetana", "Macaca_tonkeana", "Macaca_tonkeana")
st <- c(0.65, 0.94, 0.79, 0.56, 0.92, 0.65, 0.78,
        0.49, 0.60, 0.45, 0.87, 0.80, 0.22, 0.20)
specdata <- data.frame(spec = sp,
                       steepness = st)

# prune tree to 9 macaque species
species <- unique(specdata$spec)
tree <- keep.tip(Primates301_nex, species)
# plot(tree)
length(tree$tip.label) == length(species) # sanity check
#> [1] TRUE


# prepare data set to be handed over to stan
xdata <- list()
xdata$N <- nrow(specdata)
xdata$the_steepness <- specdata$steepness
# comment out/ignore the following line if fitting the original beta model
# logit transform:
xdata$the_steepness <- log(xdata$the_steepness / (1 - xdata$the_steepness))

# phylogeny-related
tree_mat <- vcv.phylo(tree, corr = TRUE) # covariance matrix
n_spec <- length(tree$tip.label)
# index for species
species_index <- integer(nrow(specdata))
for (i in seq_len(nrow(specdata))) {
  species_index[i] <- which(colnames(tree_mat) == specdata$spec[i])
}
```

[4]Replace `"comparative_model_2_gaussian.stan"` with `"comparative_model_2.stan"` to load and fit the original beta model.

```r
# for phylogeny part
xdata$N_phyl <- n_spec # number of levels
xdata$N_coeff_phyl <- 1 # number of pars: SD for phyl 'intercepts'
xdata$spec_index <- species_index
xdata$chol_mat <- t(chol(tree_mat)) # cholesky factor of phylogenetic correlation matrix
xdata$phyl_predictor <- rep(1, nrow(specdata)) # group-level predictor values

# for repeated measures part
xdata$N_repeated <- n_spec # the same as for the phyl part (number of levels)
# number of pars: SD for repeated measurements (species intercepts)
xdata$N_coeff_repeated <- 1
xdata$index_rep_measures <- species_index # the same as for the phyl part
# group-level predictor values, the same as for the phyl part
xdata$rep_measures_predictor <- rep(1, nrow(specdata))
```

```r
# sampling
res2 <- xmod2$sample(data = xdata,
                     refresh = 0,
                     parallel_chains = 2,
                     seed = 123,
                     adapt_delta = 0.99)
```

```r
# convert back to rstan (for easier handling of output)
stanfit2 <- read_stan_csv(res2$output_files())
xsummary2 <- summary(stanfit2)$summary
round(summary(stanfit2,
      pars = c("sd_phyl", "sd_repeated_measures", "sd_residual"))$summary, 2)
#>                           mean se_mean   sd 2.5%  25%  50%  75% 97.5%   n_eff
#> sd_phyl[1]                0.84    0.01 0.55 0.04 0.43 0.78 1.16  2.07 1353.41
#> sd_repeated_measures[1] 0.71    0.01 0.51 0.03 0.31 0.64 1.00  1.86 1304.20
#> sd_residual               0.96    0.01 0.30 0.55 0.75 0.91 1.12  1.68 1678.54
#>                           Rhat
#> sd_phyl[1]                   1
#> sd_repeated_measures[1]    1
#> sd_residual                  1

# optional saving of results
# save(xmod2, xsummary2, res2, stanfit2, xdata, specdata, tree, file = "comp_analysis_2.RData")

# plot in the manuscript
sd_phyl <- sprintf("%.2f", xsummary2["sd_phyl[1]", 1])
sd_rep <- sprintf("%.2f", xsummary2["sd_repeated_measures[1]", 1])
sd_resid <- sprintf("%.2f", xsummary2["sd_residual", 1])

par(mfrow = c(1, 2), family = "serif")
par(family = "serif", mgp = c(1.8, 0.7, 0), las = 1, mar = c(2, 1, 1, 1))

# plot tree with number of groups
ape::plot.phylo(tree, cex = 0.8, show.tip.label = FALSE, x.lim = c(0, 19))
tips <- tree$tip.label
for (i in seq_along(tips)) {
  x <- sum(specdata$spec == tips[i])
  y <- gsub("_", " ", tips[i])
  ape::tiplabels(text = bquote(italic(.(y))~.(paste0(" (", x, ")"))),
```

```
                tip = i, bg = NULL, adj = 0,
                frame = "none", offset = 0.3, cex = 0.8)
}
par(mar = c(3, 2.5, 1.5, 1))

# extract posterior samples for SDs and corresponding priors
est_phyl <- as.numeric(extract(stanfit2, pars = c("sd_phyl"))$sd_phyl)
est_rep <- as.numeric(extract(stanfit2, pars = c("sd_repeated_measures"))$sd_repeated_measures)
est_resid <- as.numeric(extract(stanfit2, pars = c("sd_residual"))$sd_residual)

pr_phyl <- as.numeric(extract(stanfit2, pars = c("prior_sd_phyl"))$prior_sd_phyl)
pr_rep <- as.numeric(extract(stanfit2, pars = c("prior_sd_repeated_measures"))$prior_sd_repeated_measur
pr_resid <- as.numeric(extract(stanfit2, pars = c("prior_sd_residual"))$prior_sd_residual)

# and densities of posteriors
est_phyl <- density(est_phyl, adjust = 1)
est_rep <- density(est_rep, adjust = 1)
est_resid <- density(est_resid, adjust = 1)

# and for priors
pr_phyl <- density(pr_phyl[pr_phyl < 10], adjust = 1) # truncated for smoother display
pr_rep <- density(pr_rep[pr_rep < 10], adjust = 1)
pr_resid <- density(pr_resid[pr_resid < 10], adjust = 1)


# data ranges for plot
yr <- c(0, max(est_phyl$y, est_rep$y, est_resid$y) * 1.05)
xr <- c(0, max(est_phyl$x, est_rep$x, est_resid$x) * 1.05)


plot(0, 0, type = "n", xlim = xr, ylim = yr, xaxs = "i", yaxs = "i",
     las = 1, axes = FALSE, xlab = "SD", ylab = "density")
axis(1)

polygon(c(est_phyl$x, rev(est_phyl$x)), c(rep(0, length(est_phyl$y)), rev(est_phyl$y)),
        border = NA, col = adjustcolor("#3B99B1", 0.5))
polygon(c(est_rep$x, rev(est_rep$x)), c(rep(0, length(est_rep$y)), rev(est_rep$y)),
        border = NA, col = adjustcolor("#EACB2B", 0.5))
polygon(c(est_resid$x, rev(est_resid$x)), c(rep(0, length(est_resid$y)), rev(est_resid$y)),
        border = NA, col = adjustcolor("#F5191C", 0.5))

polygon(c(pr_phyl$x, rev(pr_phyl$x)), c(rep(0, length(pr_phyl$y)), rev(pr_phyl$y)),
        border = NA, col = adjustcolor("grey", 0.4))
polygon(c(pr_rep$x, rev(pr_rep$x)), c(rep(0, length(pr_rep$y)), rev(pr_rep$y)),
        border = NA, col = adjustcolor("grey", 0.4))
polygon(c(pr_resid$x, rev(pr_resid$x)), c(rep(0, length(pr_resid$y)), rev(pr_resid$y)),
        border = NA, col = adjustcolor("grey", 0.4))

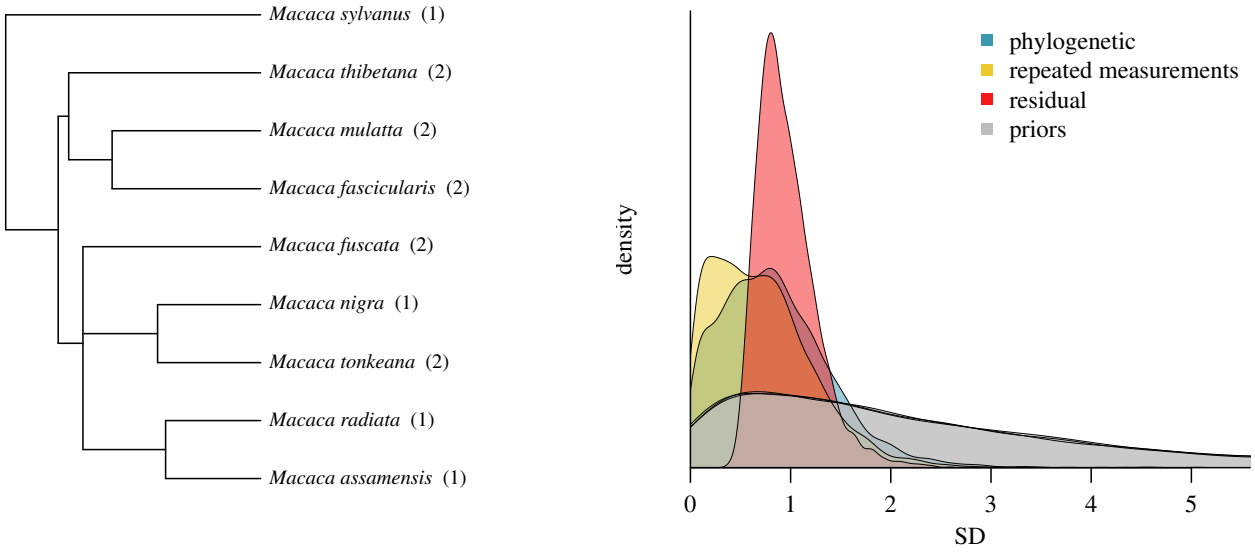
points(est_phyl$x, est_phyl$y, type = "l", lwd = 0.5)
points(est_rep$x, est_rep$y, type = "l", lwd = 0.5)
points(est_resid$x, est_resid$y, type = "l", lwd = 0.5)

points(pr_phyl$x, pr_phyl$y, type = "l", lwd = 0.5)
```

```
points(pr_rep$x, pr_rep$y, type = "l", lwd = 0.5)
points(pr_resid$x, pr_resid$y, type = "l", lwd = 0.5)

box(bty = "l")

legend("topright", ncol = 1, col = c("#3B99B1", "#EACB2B", "#F5191C", "grey"),
       legend = c("phylogenetic", "repeated measurements", "residual", "priors"),
       pch = 15, bty = "n")
```



And some posterior predictive checks. First we compare the observed steepness to our model's predictions using the logit-transformed values, i.e. we stay on the scale in which the modelling happened. The second plot back-transforms steepness to its original scale, but note that these transformations happened *after* the modelling.

```
set.seed(123)
# the density of the observed steepnesses
# (logit of steepness reported in original paper)
d_observed <- density(xdata$the_steepness, adjust = 1)

# draws of modeled predictions for steepness
d <- extract(stanfit2, "steepness_rep")$steepness_rep

plot(0, 0, xlim = c(-2, 5), ylim = c(0, 0.8), type = "n", xaxs = "i", yaxs = "i",
     xlab = "steepness (logit scale)", ylab = "density", axes = FALSE)
axis(1)

for (i in 1:50) {
  p <- d[sample(nrow(d), 1), ]
  p <- density(p, adjust = 1.5)
  points(p$x, p$y, type = "l", lwd = 0.5)
}

# add observed
points(d_observed$x, d_observed$y, type = "l", lwd = 3, col = "red")
box(bty = "l")
```

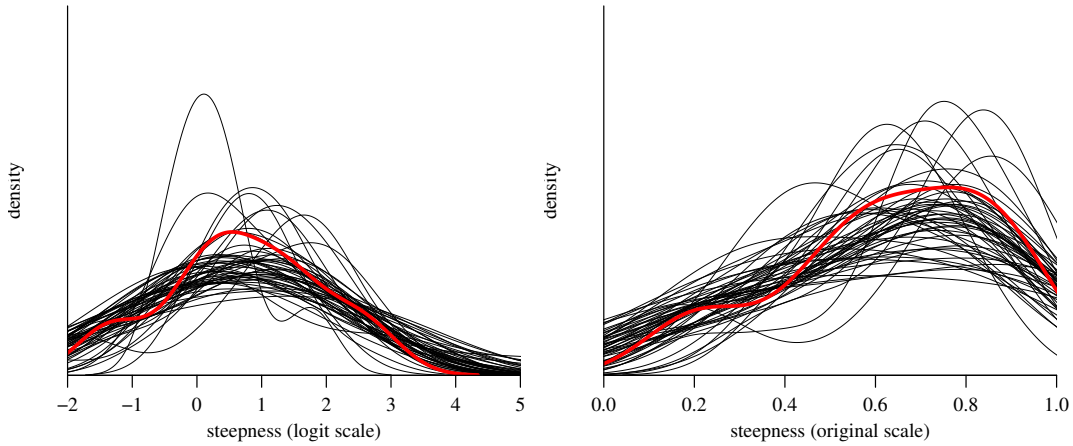
```r
# and the same on the original scale
d_observed <- density(specdata$steepness, adjust = 1)

plot(0, 0, xlim = c(0, 1), ylim = c(0, 3), type = "n", xaxs = "i", yaxs = "i",
     xlab = "steepness (original scale)", ylab = "density", axes = FALSE)
axis(1)

inv_logit <- function(x) exp(x) / (1 + exp(x))
for (i in 1:50) {
  # need to inverse logit posterior samples
  p <- inv_logit(d[sample(nrow(d), 1), ])

  p <- density(p, adjust = 1.5)
  points(p$x, p$y, type = "l", lwd = 0.5)
}

points(d_observed$x, d_observed$y, type = "l", lwd = 3, col = "red")
box(bty = "l")
```



Finally, this latter model can also be fit using `brms`.

```r
specdata <- data.frame(spec_phyl = sp,
                       spec_rep = sp,
                       steepness = st)
# logit transform
specdata$steepness_transformed <- log(specdata$steepness / (1 - specdata$steepness))

maca_phylo <- ape::vcv.phylo(tree, corr = TRUE)
pres <- brm(steepness_transformed ~ 1 +
              (1|gr(spec_phyl, cov = maca_phylo)) +
              (1|spec_rep),
            data = specdata,
            data2 = list(maca_phylo = maca_phylo),
            family = gaussian(),
            sample_prior = TRUE,
            cores = 2,
            backend = "cmdstanr",
            control = list(adapt_delta = 0.99))
```

```
pres
```

```
#>  Family: gaussian
#>   Links: mu = identity; sigma = identity
#> Formula: steepness_transformed ~ 1 + (1 | gr(spec_phyl, cov = maca_phylo)) + (1 | spec_rep)
#>    Data: specdata (Number of observations: 14)
#>   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
#>          total post-warmup draws = 4000
#>
#> Group-Level Effects:
#> ~spec_phyl (Number of levels: 9)
#>               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
#> sd(Intercept)     0.83      0.55     0.03     2.09 1.00     1202     1504
#>
#> ~spec_rep (Number of levels: 9)
#>               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
#> sd(Intercept)     0.72      0.51     0.03     1.91 1.00     1206     1839
#>
#> Population-Level Effects:
#>           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
#> Intercept     0.64      0.61    -0.55     1.87 1.00     2096     2392
#>
#> Family Specific Parameters:
#>       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
#> sigma     0.97      0.30     0.54     1.71 1.00     1890     2604
#>
#> Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
#> and Tail_ESS are effective sample size measures, and Rhat is the potential
#> scale reduction factor on split chains (at convergence, Rhat = 1).
```

3 Data summary

Table 1 provides data summaries and references for the analyses run above.

Table 1: Macaque dominance interaction networks

data set	species	common name	group size	interactions	sparseness	reference
anderson2016b_dom_1	*Macaca fuscata*	Japanese macaque	6	80	0.20	Anderson et al. (2016); Anderson (2016)
anderson2016b_dom_2	*Macaca fuscata*	Japanese macaque	5	28	0.30	Anderson et al. (2016); Anderson (2016)
balasubramaniam2014_dom_1	*Macaca tonkeana*	Tonkean macaque	9	174	0.14	Balasubramaniam (2014); Demaria & Thierry (2001)
balasubramaniam2014_dom_2	*Macaca nigra*	crested macaque	6	516	0.00	Balasubramaniam (2014)
bartlett1971_dom_1	*Macaca mulatta*	rhesus macaque	9	37	0.31	Bartlett & Meier (1971)
bartlett1971_dom_2	*Macaca mulatta*	rhesus macaque	9	90	0.06	Bartlett & Meier (1971)
berman2004_dom_1	*Macaca thibetana*	Tibetan macaque	13	263	0.09	Berman et al. (2004)
berman2004_dom_2	*Macaca thibetana*	Tibetan macaque	13	379	0.20	Berman et al. (2004)
berman2004_dom_3	*Macaca thibetana*	Tibetan macaque	10	428	0.02	Berman et al. (2004)
birky2002_dom_01	*Macaca cyclopis*	Formosan macaque	6	109	0.00	Birky (2002)
birky2002_dom_02	*Macaca cyclopis*	Formosan macaque	6	61	0.07	Birky (2002)
birky2002_dom_12	*Macaca cyclopis*	Formosan macaque	6	66	0.07	Birky (2002)
chacko2014_dom_1	*Macaca radiata*	bonnet macaque	11	354	0.04	Chacko (2014)
chapais1985_dom_1	*Macaca fuscata*	Japanese macaque	10	97	0.40	Chapais (1985)
cooper1999_dom_1	*Macaca assamensis*	Assamese macaque	22	1077	0.16	Cooper (1999)
deng1987_dom_1	*Macaca thibetana*	Tibetan macaque	15	383	0.26	Deng & Zhao (1987)
dewaal1977_dom_1	*Macaca fascicularis*	longtailed macaque	6	102	0.20	de Waal (1977)
dewaal1977_dom_2	*Macaca fascicularis*	longtailed macaque	11	542	0.09	de Waal (1977)
dewaal1985_dom_1	*Macaca mulatta*	rhesus macaque	20	856	0.10	de Waal & Luttrell (1985)
duboscq2013_dom_1	*Macaca nigra*	crested macaque	15	561	0.04	Duboscq et al. (2013)
duboscq2013_dom_2	*Macaca nigra*	crested macaque	15	210	0.23	Duboscq et al. (2013)
duboscq2013_dom_3	*Macaca nigra*	crested macaque	15	407	0.10	Duboscq et al. (2013)
duboscq2013_dom_4	*Macaca nigra*	crested macaque	21	283	0.36	Duboscq et al. (2013)
duboscq2013_dom_5	*Macaca nigra*	crested macaque	21	732	0.11	Duboscq et al. (2013)
duboscq2013_dom_6	*Macaca nigra*	crested macaque	21	711	0.12	Duboscq et al. (2013)
fa1986_dom_1	*Macaca sylvanus*	Barbary macaque	6	70	0.00	Fa (1986)
fa1986_dom_2	*Macaca sylvanus*	Barbary macaque	6	43	0.20	Fa (1986)
fa1986_dom_3	*Macaca sylvanus*	Barbary macaque	6	51	0.07	Fa (1986)
fa1986_dom_4	*Macaca sylvanus*	Barbary macaque	6	38	0.13	Fa (1986)
funkhouser2018a_dom_5	*Macaca thibetana*	Tibetan macaque	13	53	0.63	Funkhouser et al. (2018)
funkhouser2018a_dom_6	*Macaca thibetana*	Tibetan macaque	13	69	0.49	Funkhouser et al. (2018)
furuichi1983_dom_1	*Macaca fuscata*	Japanese macaque	7	15	0.48	Furuichi (1983)
furuichi1983_dom_2	*Macaca fuscata*	Japanese macaque	7	21	0.38	Furuichi (1983)
giacoma1992_dom_1	*Macaca nemestrina*	pigtailed macaque	6	1105	0.00	Giacoma & Messeri (1992)
giacoma1992_dom_2	*Macaca nemestrina*	pigtailed macaque	6	287	0.00	Giacoma & Messeri (1992)
giacoma1992_dom_3	*Macaca nemestrina*	pigtailed macaque	6	419	0.00	Giacoma & Messeri (1992)
gumert2000_dom_1	*Macaca fascicularis*	longtailed macaque	10	16	0.73	Gumert (2000)
gumert2000_dom_3	*Macaca fascicularis*	longtailed macaque	10	35	0.73	Gumert (2000)
hanya2008_dom_1	*Macaca fuscata*	Japanese macaque	9	96	0.08	Hanya et al. (2008)
hanya2008_dom_2	*Macaca fuscata*	Japanese macaque	7	22	0.38	Hanya et al. (2008)
hill1995_dom_1	*Macaca fuscata*	Japanese macaque	7	291	0.00	Hill & Okayasu (1995)
kaburu2012_dom_1	*Macaca sylvanus*	Barbary macaque	7	584	0.00	Kaburu et al. (2012)
matsumura1998_dom_1	*Macaca maura*	moor macaque	10	112	0.29	Matsumura (1998)
micheletta2011_dom_1	*Macaca nigra*	crested macaque	6	277	0.07	Micheletta (pers. comm.)
missakian1972_dom_1	*Macaca mulatta*	rhesus macaque	22	5159	0.00	Missakian (1972)
murray1984_dom_1	*Macaca arctoides*	stumptailed macaque	20	1176	0.06	Murray (1984)
oi1988_dom_1	*Macaca fuscata*	Japanese macaque	16	45	0.69	Oi (1988)
oi1988_dom_2	*Macaca fuscata*	Japanese macaque	5	3	0.70	Oi (1988)
oi1988_dom_3	*Macaca fuscata*	Japanese macaque	10	85	0.27	Oi (1988)
oi1990_dom_1	*Macaca nemestrina*	pigtailed macaque	15	359	0.17	Oi (1990)
petit1992_dom_1	*Macaca maura*	moor macaque	5	70	0.30	Petit & Thierry (1992)
petit1992_dom_2	*Macaca maura*	moor macaque	5	231	0.00	Petit & Thierry (1992)
raney1981_dom_1	*Macaca radiata*	bonnet macaque	24	276	0.54	Raney et al. (1981)
rhine1972_dom_1	*Macaca arctoides*	stumptailed macaque	4	154	0.00	Rhine (1972); Rhine & Kronenwetter (1972)

Continued on next page

Table 1: Macaque dominance interaction networks

data set	species	common name	group size	interactions	sparseness	reference
rhine1972_dom_3	*Macaca arctoides*	stumptailed macaque	4	204	0.00	Rhine (1972); Rhine & Kronenwetter (1972)
rizaldi2010_dom_2	*Macaca fuscata*	Japanese macaque	5	81	0.00	Rizaldi & Watanabe (2010)
sade1972_dom_1	*Macaca mulatta*	rhesus macaque	9	136	0.00	Sade (1972)
scanlon1984_dom_1	*Macaca mulatta*	rhesus macaque	9	116	0.08	Scanlon (1984)
silk1981_dom_1	*Macaca radiata*	bonnet macaque	29	1413	0.25	Silk et al. (1981)
sterck1997_dom_01	*Macaca fascicularis*	longtailed macaque	9	15	0.69	Sterck & Steenbeek (1997)
sterck1997_dom_02	*Macaca fascicularis*	longtailed macaque	9	36	0.42	Sterck & Steenbeek (1997)
su2003_dom_1	*Macaca cyclopis*	Formosan macaque	8	245	0.04	Su (2003)
su2003_dom_2	*Macaca cyclopis*	Formosan macaque	8	42	0.25	Su (2003)
tully1975_dom_1	*Macaca radiata*	bonnet macaque	8	247	0.00	Tully (1975)
tully1975_dom_2	*Macaca nemestrina*	pigtailed macaque	8	100	0.11	Tully (1975)
zhang2014b_dom_1	*Macaca thibetana*	Tibetan macaque	9	288	0.00	Zhang et al. (2014)
zumpe1986_dom_1	*Macaca mulatta*	rhesus macaque	4	55	0.00	Zumpe & Michael (1986)

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